第6章　非正弦信号与频率特性分析

(*nonsinusoidal*) (*Frequency characteristics*)

6.1 周期非正弦信号的分解与频谱

6.2 周期非正弦信号的有效值和平均功率

6.3 非正弦电路的稳态分析

6.4* 非周期信号与傅里叶变换概念

6.5 电路的频率特性分析

6.6* 对称三相电路中的高次谐波

6.7* 频率响应 、滤波器

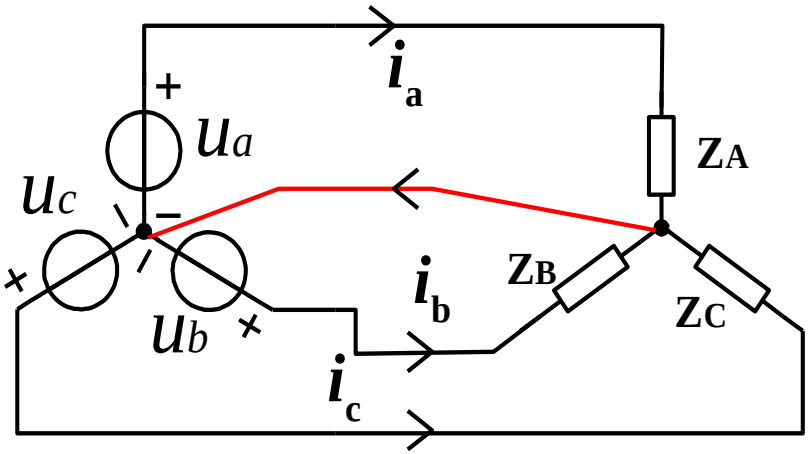
6.6* 对称三相电路中的高次谐波

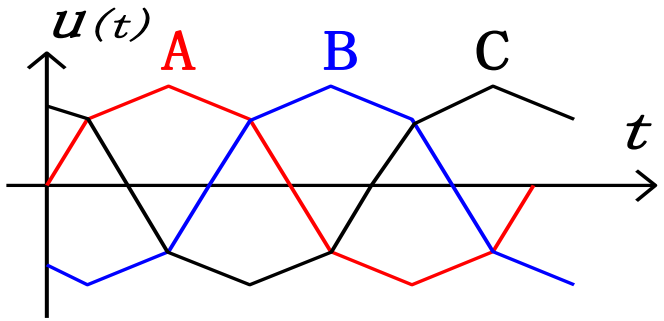
在实际的电力系统中，三相发电机产生的电压往往不是理想的正弦波。电网中变压器等设备由于磁路的非线性，其励磁电流往往是非正弦周期波形，包含有高次谐波分量。因此在三相对称电路中，电网电压与电流都可能产生非正弦波形，即存在高次谐波。

1. 三相对称非正弦电压的分析

三相对称非正弦电压

$$\begin{cases} u_A = u(t) \\ u_B = u(t-\dfrac{T}{3}) \\ u_C = u(t-\dfrac{2}{3}T) \end{cases}$$





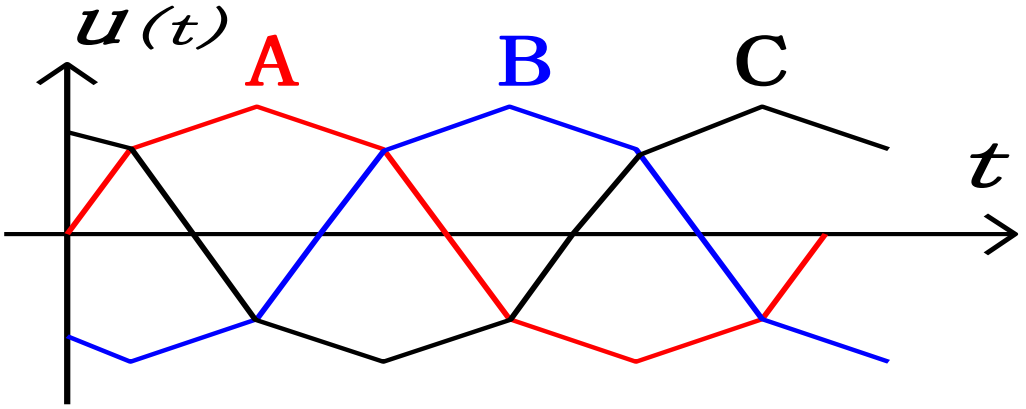
$$u_A(t)=\sqrt{2}\,E_1\sin(\omega t+\psi_1)+\sqrt{2}\,E_3\sin(3\omega t+\psi_3)$$
$$+\sqrt{2}\,E_5\sin(5\omega t+\psi_5)+\sqrt{2}\,E_7\sin(7\omega t+\psi_7)+\cdots$$

$$u_B(t)=\sqrt{2}\,E_1\sin\left[\omega\left(t-\frac{T}{3}\right)+\psi_1\right]+\sqrt{2}\,E_3\sin\left[3\omega\left(t-\frac{T}{3}\right)+\psi_3\right]$$
$$+\sqrt{2}\,E_5\sin\left[5\omega\left(t-\frac{T}{3}\right)+\psi_5\right]+\sqrt{2}\,E_7\sin\left[7\omega\left(t-\frac{T}{3}\right)+\psi_7\right]+\cdots$$

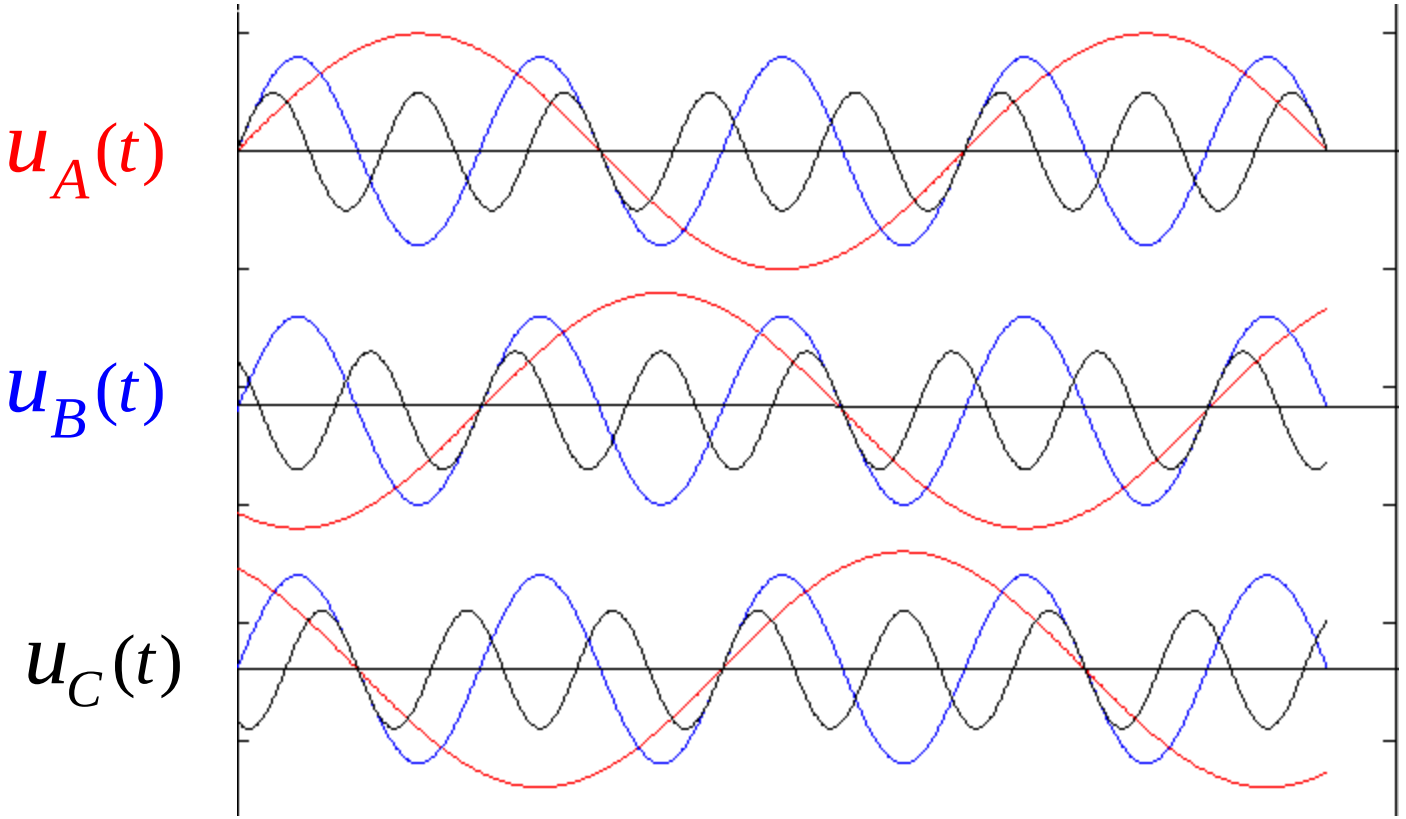
即
$$u_B(t)=\sqrt{2}\,E_1\sin\left(\omega t+\psi_1-\frac{2}{3}\pi\right)+\sqrt{2}\,E_3\sin[3\omega t+\psi_3]$$
$$+\sqrt{2}\,E_5\sin\left[5\omega t+\psi_5-\frac{4}{3}\pi\right]+\sqrt{2}\,E_7\sin\left[7\omega t+\psi_7-\frac{2}{3}\pi\right]+\cdots$$

同理有
$$u_C(t)=\sqrt{2}\,E_1\sin\left(\omega t+\psi_1-\frac{4}{3}\pi\right)+\sqrt{2}\,E_3\sin[3\omega t+\psi_3]$$
$$+\sqrt{2}\,E_5\sin\left[5\omega t+\psi_5-\frac{2}{3}\pi\right]+\sqrt{2}\,E_7\sin\left[7\omega t+\psi_7-\frac{4}{3}\pi\right]+\cdots$$

三相对称非正弦电压



三相对称非正弦
电压分解波形



$n=6k+1$ $(k=0,1,2,\cdots)$次谐波分量 (基波,7次谐波等)

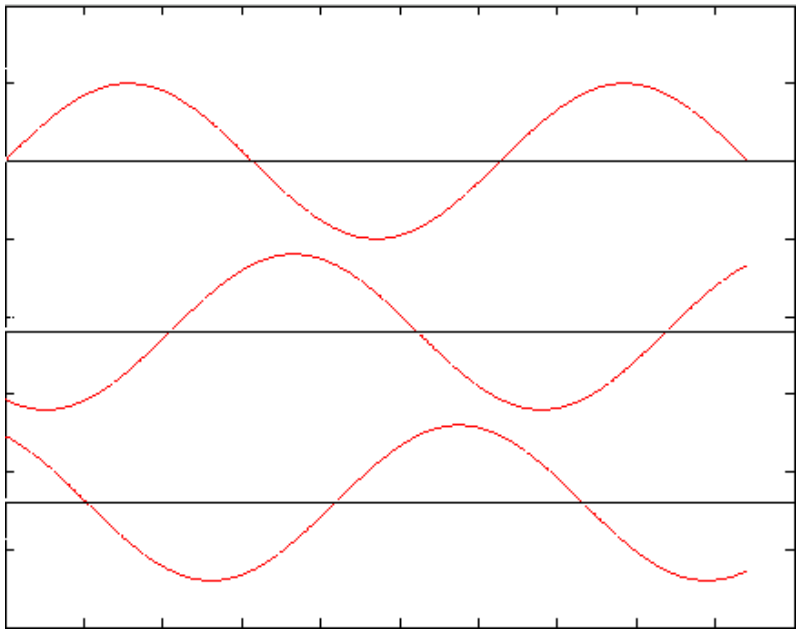
1. 对称三相正序系统

$$u_{A1}(t)=\sqrt{2}\,E_1\sin(\omega t+\psi_1)$$

$$u_{B1}(t)=\sqrt{2}\,E_1\sin\left(\omega t+\psi_1-\frac{2}{3}\pi\right)$$

$$u_{C1}(t)=\sqrt{2}\,E_1\sin\left(\omega t+\psi_1-\frac{4}{3}\pi\right)$$

基波



相序变化依次为

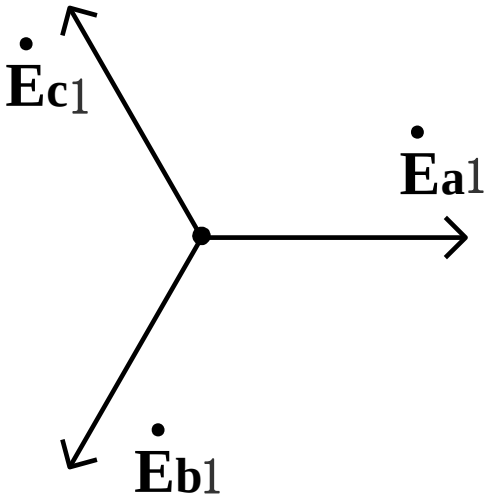
A→B→C→A

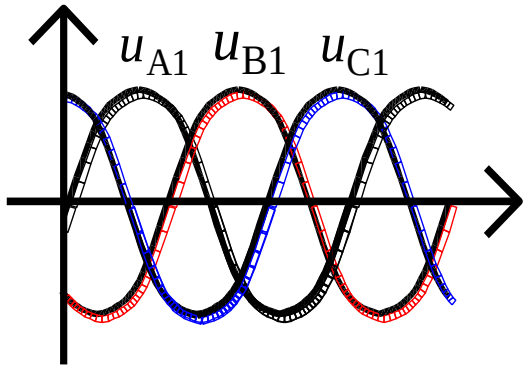
$n = 6k + 1$ $(k = 0, 1, 2, \cdots)$次谐波分量 (基波,7次谐波等)

$$u_{A1}(t) = \sqrt{2}\, E_1 \sin(\omega t + \psi_1)$$

$$u_{B1}(t) = \sqrt{2}\, E_1 \sin\left(\omega t + \psi_1 - \frac{2}{3}\pi\right)$$

$$u_{C1}(t) = \sqrt{2}\, E_1 \sin\left(\omega t + \psi_1 - \frac{4}{3}\pi\right)$$





对称三相正序系统的计算：化为单相计算.

2. 对称三相负序系统

$n = 6k - 1 \ (k = 1, 2, \cdots)$ 次谐波分量 (5次谐波等)

$$u_{A5}(t) = \sqrt{2}\, E_5 \sin(5\omega t + \psi_5)$$

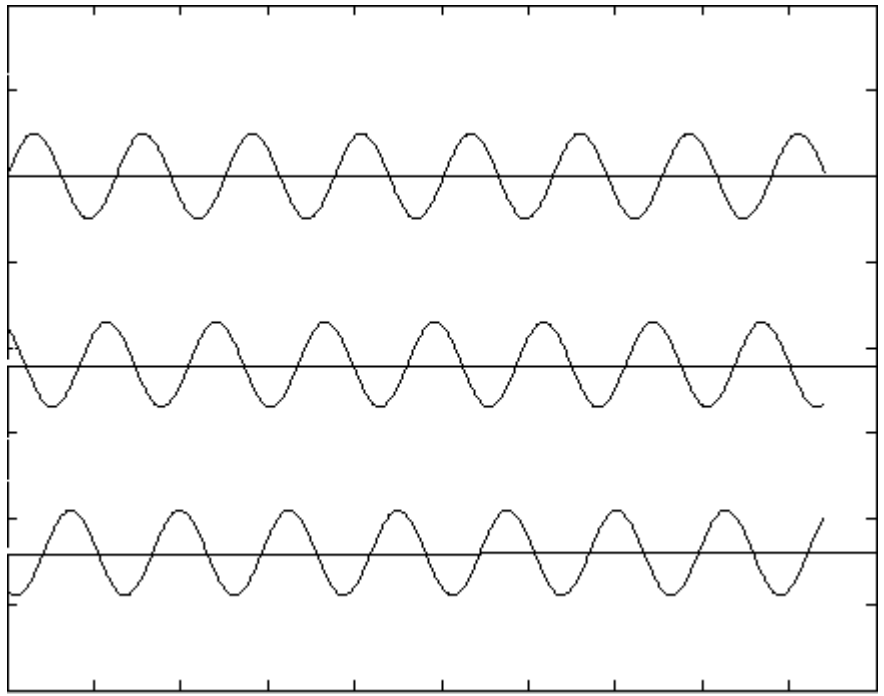
$$u_{B5}(t) = \sqrt{2}\, E_5 \sin\left(5\omega t + \psi_5 + \frac{2}{3}\pi\right)$$

$$u_{C5}(t) = \sqrt{2}\, E_5 \sin\left(5\omega t + \psi_5 - \frac{2}{3}\pi\right)$$

5次谐波

相序变化依次为

A→C→B→A

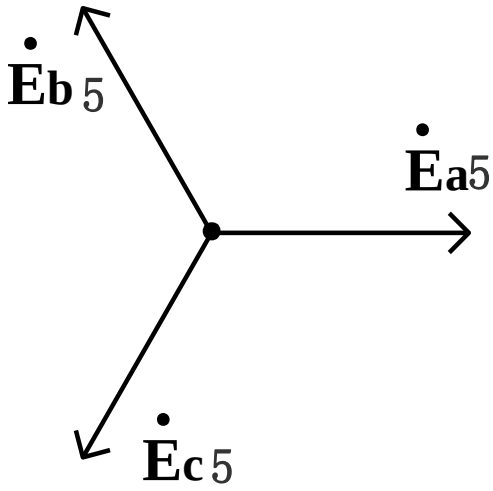


$n=6k-1\ (k=1,2,\cdots)$ 次谐波分量 (5次谐波等)

$$u_{A5}(t)=\sqrt{2}\,E_5\sin(5\omega t+\psi_5)$$

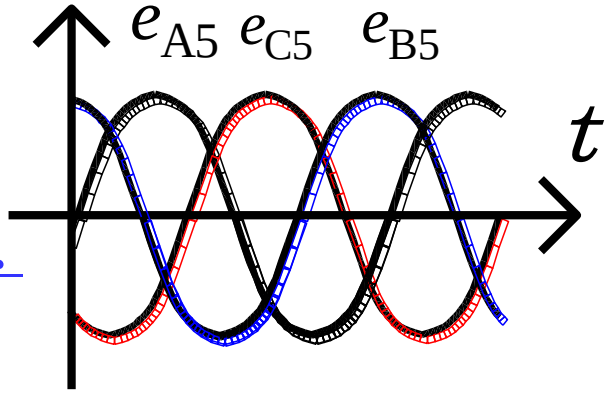
$$u_{B5}(t)=\sqrt{2}\,E_5\sin\left(5\omega t+\psi_5+\frac{2}{3}\pi\right)$$

$$u_{C5}(t)=\sqrt{2}\,E_5\sin\left(5\omega t+\psi_5-\frac{2}{3}\pi\right)$$



对称三相负序系统的计算:单相图.

写其余两相时注意相序与正序系统的区别.



3. 对称三相零序系统

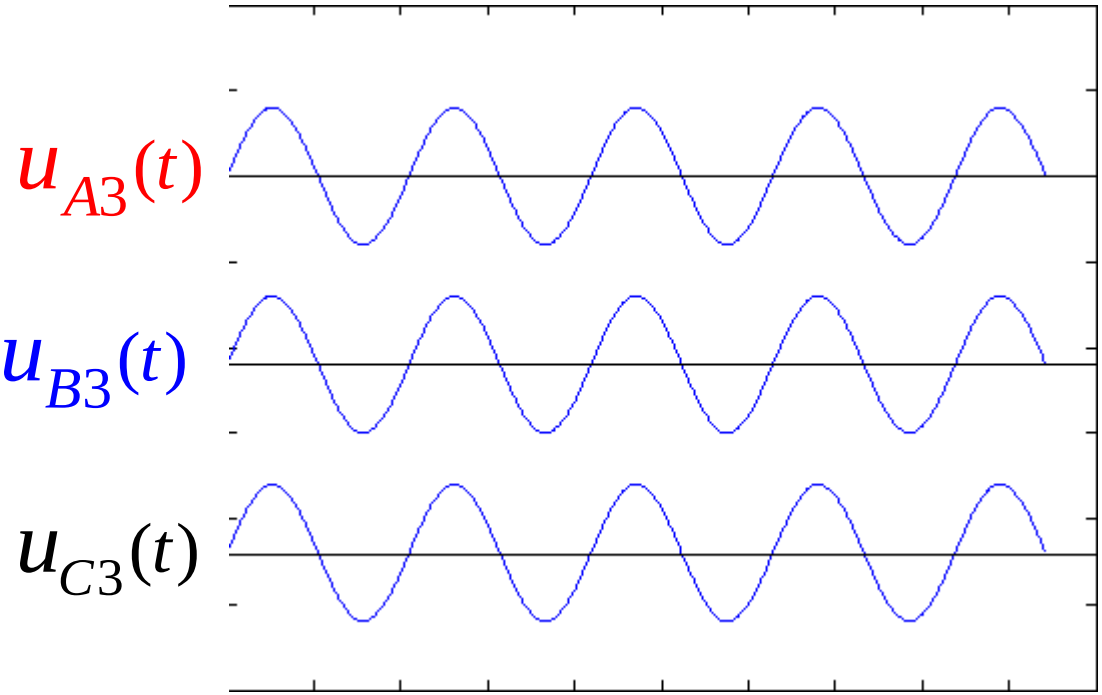
$n=3k+3\ (k=1,2,\cdots)$ 次谐波分量 (3次谐波等)

$$u_{A3}(t)=\sqrt{2}\,E_3\sin(3\omega t+\psi_3)$$

$$u_{B3}(t)=\sqrt{2}\,E_3\sin\left(3\omega t+\psi_3\right)$$

$$u_{C3}(t)=\sqrt{2}\,E_3\sin\left(3\omega t+\psi_3\right)$$

3次谐波



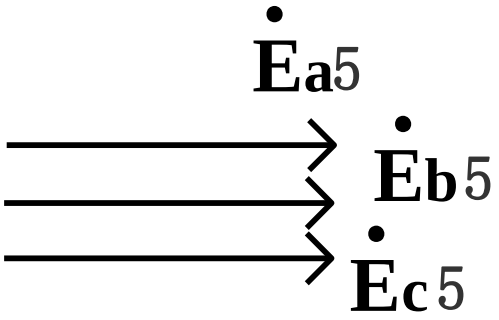
相序变化: 各相分量
振幅相等、相位相同

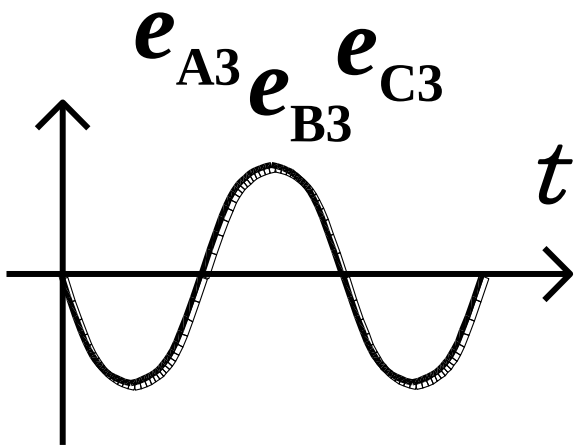
$n=3k+3\ (k=1,2,\cdots)$ 次谐波分量 (3次谐波等)

$$u_{A3}(t)=\sqrt{2}\,E_3\sin(3\omega t+\psi_3)$$

$$u_{B3}(t)=\sqrt{2}\,E_3\sin\left(3\omega t+\psi_3\right)$$

$$u_{C3}(t)=\sqrt{2}\,E_3\sin\left(3\omega t+\psi_3\right)$$

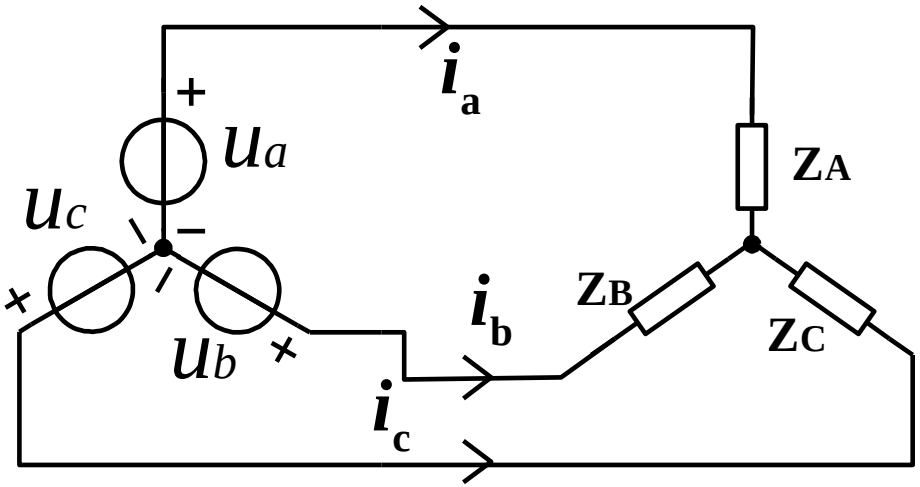




对称三相零序系统的计算根据电路的不同分别讨论.

1>. Y-Y无中线系统

正序和负序系统的各谐波分量，线电压有效值是对应相电压分量有效值的$\sqrt{3}$倍，零序分量由于幅值相等相位相同，因此在线电压中将不包含这些谐波分量。



电源相电压有效值：$U_{\mathrm{ph}}=\sqrt{U_{1\mathrm{ph}}^2+U_{3\mathrm{ph}}^2+U_{5\mathrm{ph}}^2+U_{7\mathrm{ph}}^2+\cdots}$

线电压有效值：$U_1=\sqrt{U_{1l}^2+U_{5l}^2+U_{7l}^2+U_{11l}^2+\cdots}$

$$=\sqrt{3}\times\sqrt{U_{1\mathrm{ph}}^2+U_{5\mathrm{ph}}^2+U_{7\mathrm{ph}}^2+U_{11\mathrm{ph}}^2+\cdots}$$

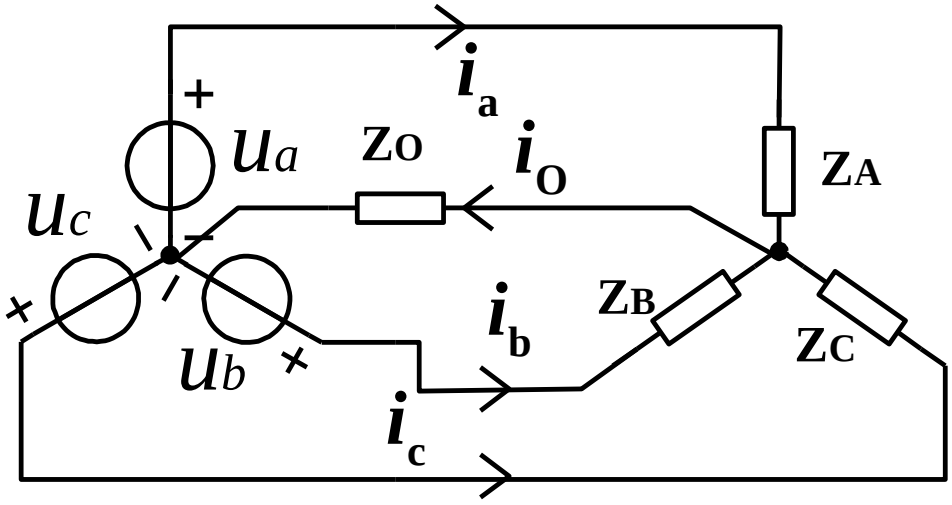
三相负载中无三次谐波电流, 因此负载相电压也没有三次谐波分量.

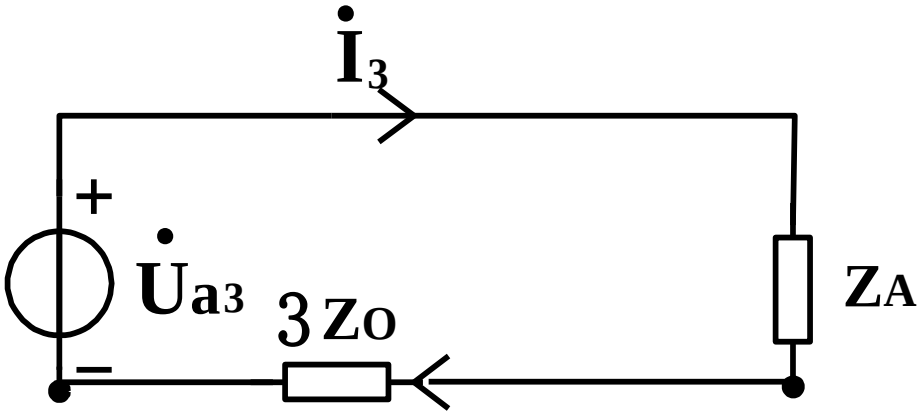
2>. Y_o-Y_o有中线系统

- 正序分量计算时,采用单相图,中线不起作用.

- 负序分量计算时,采用单相图,中线不起作用.

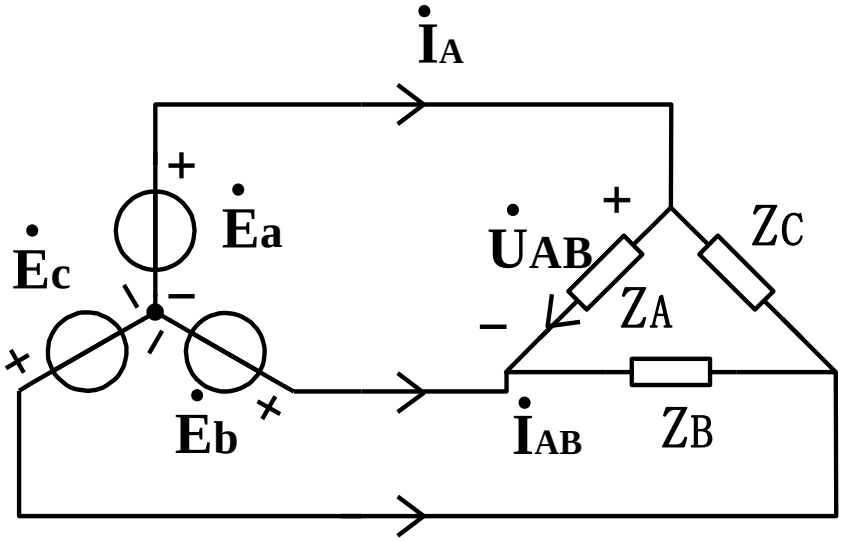
- 零序分量计算时,由于中线的存在,单相图中加入一个三倍的中线电阻.





3>. Y-Δ系统

相电压等于线电压, 负载中无三次谐波分量.

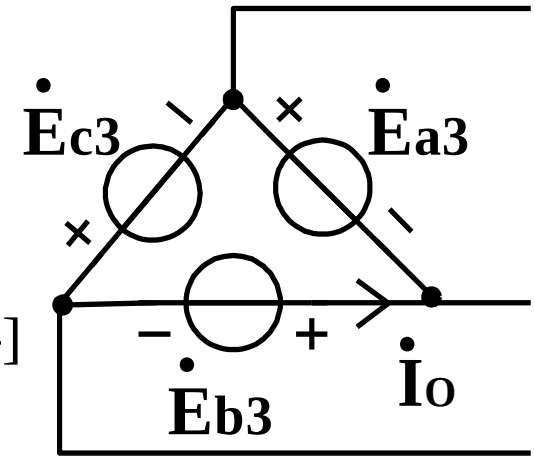


4>. Δ联结电源系统

△联结的环路中存在电动势

$$u(t)=u_{\mathrm{A}}(t)+u_{\mathrm{B}}(t)+u_{\mathrm{C}}(t)$$

$$=3[\sqrt{2}\,U_3\sin 3\omega t+\sqrt{2}\,U_9\sin(9\omega t+\psi_a)+\cdots]$$



其有效值为 $U=3\sqrt{U_3^2+U_9^2+\cdots}$

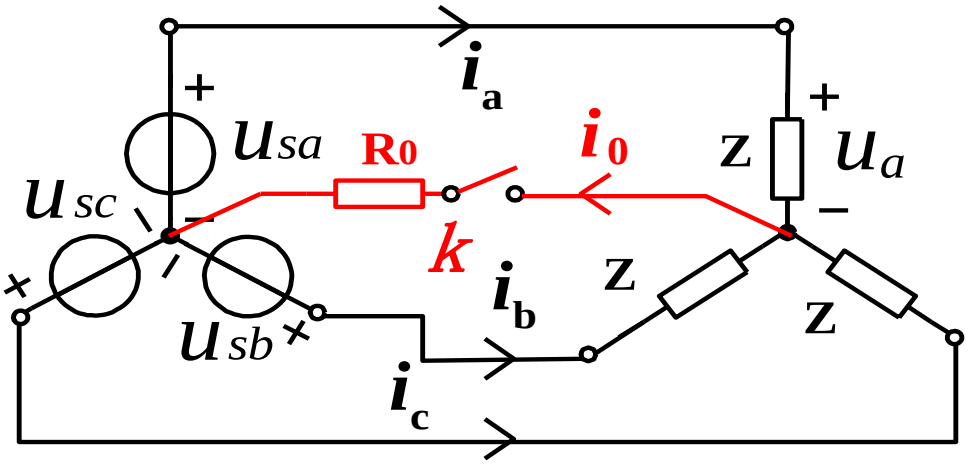
零序谐波分量会在环路中产生一个很大的谐波电流。

例1. 图示对称三相电路 $Z=10\Omega$，$R_0=5\Omega$

$$u_{sa}=100\sqrt{2}\sin(\omega t)+50\sqrt{2}\sin 3\omega t+50\sqrt{2}\sin 5\omega t$$

求：1)开关打开时电源电压有效值; 负载端的相电压,相电流和线电压的有效值;

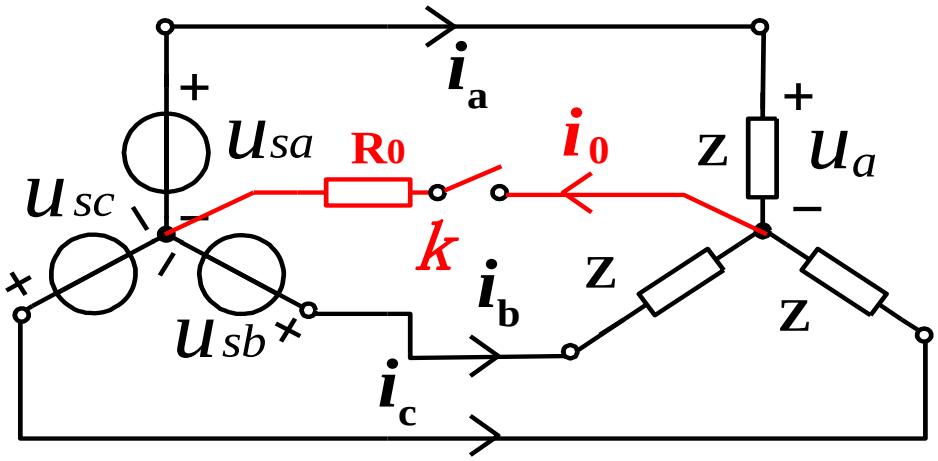
2) 开关闭合时负载端的相电压,相电流和线电压的有效值.



$$u_{sa}=100\sqrt{2}\sin(\omega t)$$

$$+50\sqrt{2}\sin 3\omega t+50\sqrt{2}\sin 5\omega t$$

$$Z=10\Omega\ ,\ R_0=5\Omega$$



开关打开时：

电源相电压：

$$U_{\mathrm{a}}=\sqrt{U_{1\mathrm{a}}^2+U_{3\mathrm{a}}^2+U_{5\mathrm{a}}^2}=\sqrt{100^2+50^2+50^2}=50\sqrt{6}$$

负载相电压：$U_{\mathrm{ph}}=\sqrt{U_{1\mathrm{ph}}^2+U_{5\mathrm{ph}}^2}=\sqrt{100^2+50^2}=50\sqrt{5}$

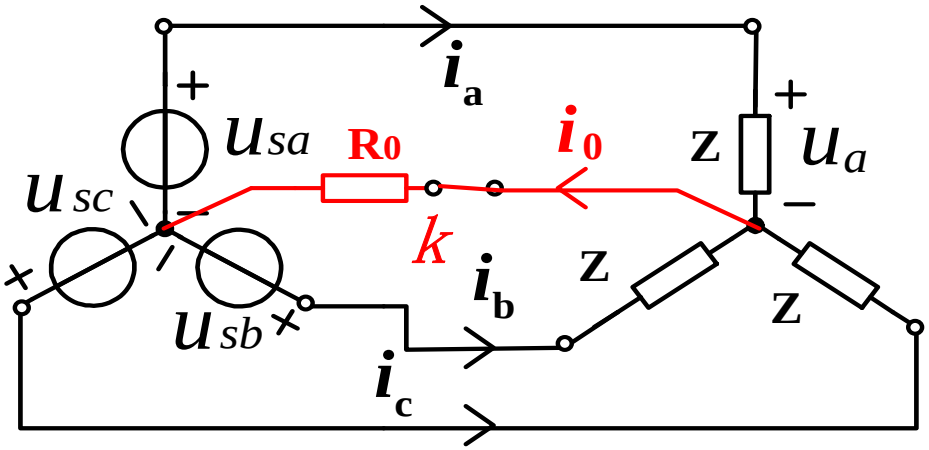
负载相电流：$I_{\mathrm{ph}}=\sqrt{I_{1\mathrm{ph}}^2+I_{5\mathrm{ph}}^2}=\sqrt{10^2+5^2}=5\sqrt{5}$

线电压：$U_{AB}=\sqrt{3}\times\sqrt{U_{1\mathrm{ph}}^2+U_{5\mathrm{ph}}^2}=50\sqrt{15}$

$$u_{sa} = 100\sqrt{2}\sin(\omega t)$$

$$+50\sqrt{2}\sin 3\omega t + 50\sqrt{2}\sin 5\omega t$$

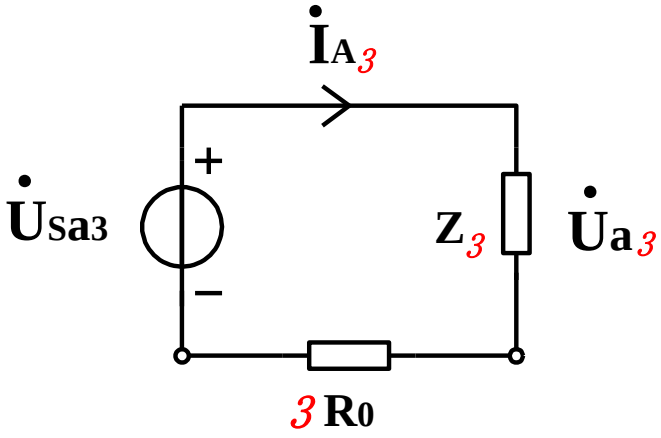
$$Z = 10\Omega\ ,\ R_0 = 5\Omega$$



开关闭合时，三次谐波:

$$\dot{I}_{a3} = \frac{\dot{U}_{sa3}}{Z_3 + 3R_0} = \frac{50\angle 0^0}{25} = 2\angle 0^0$$

$$\dot{U}_{a3} = \dot{I}_{a3} \times Z_3 = 20\angle 0^0$$

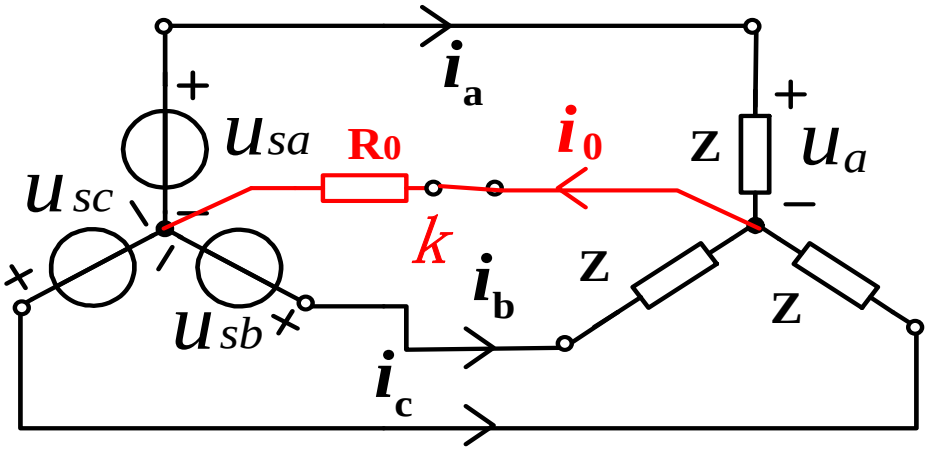


三次波单相图

$$u_{sa}=100\sqrt{2}\sin(\omega t)$$

$$+50\sqrt{2}\sin 3\omega t+50\sqrt{2}\sin 5\omega t$$

$$Z=10\Omega\ ,\ R_0=5\Omega$$



开关闭合时，三次谐波:

负载相电压:

$$U_{\text{ph}}=\sqrt{U_{1\text{ph}}^2+U_{3\text{ph}}^2+U_{5\text{ph}}^2}=\sqrt{100^2+20^2+50^2}=113.6V$$

负载相电流:

$$I_{\text{ph}}=\sqrt{I_{1\text{ph}}^2+I_{3\text{ph}}^2+I_{5\text{ph}}^2}=\sqrt{10^2+2^2+5^2}=\sqrt{129}$$

线电压:

$$U_{AB}=\sqrt{3}\times\sqrt{U_{1\text{ph}}^2+U_{5\text{ph}}^2}=50\sqrt{15}$$

问题讨论

- **为什么发电机电势中一般含有1、3、5次谐波**

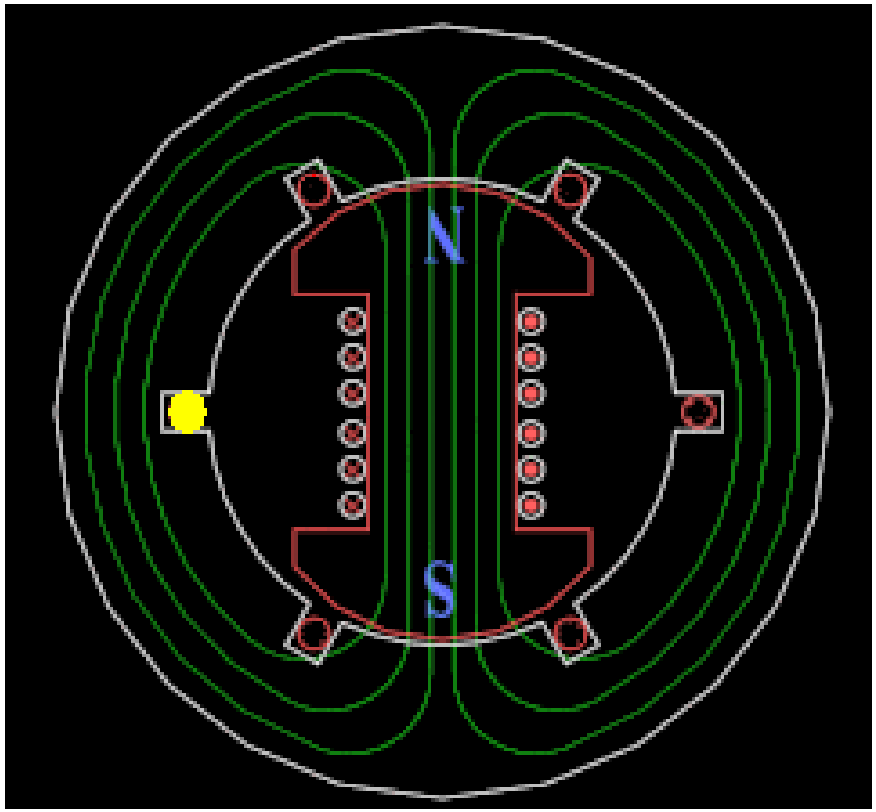
单根导体感应电动势分析

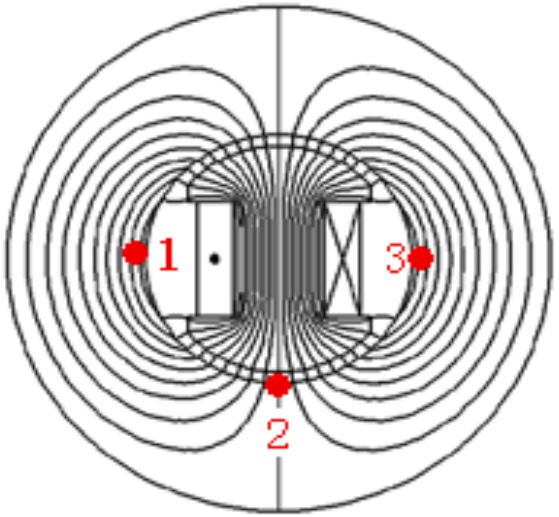
$$e = b_\delta lv$$

$$e_{A1} = B_{\delta 1m} lv \sin \omega t$$

$$= E_{1m} \sin \omega t = \sqrt{2} E_1 \sin \omega t$$

$$e_A = (B_{\delta 1m} \sin \omega t + B_{\delta 3m} \sin 3\omega t + B_{\delta 5m} \sin 5\omega t \cdots) lv$$

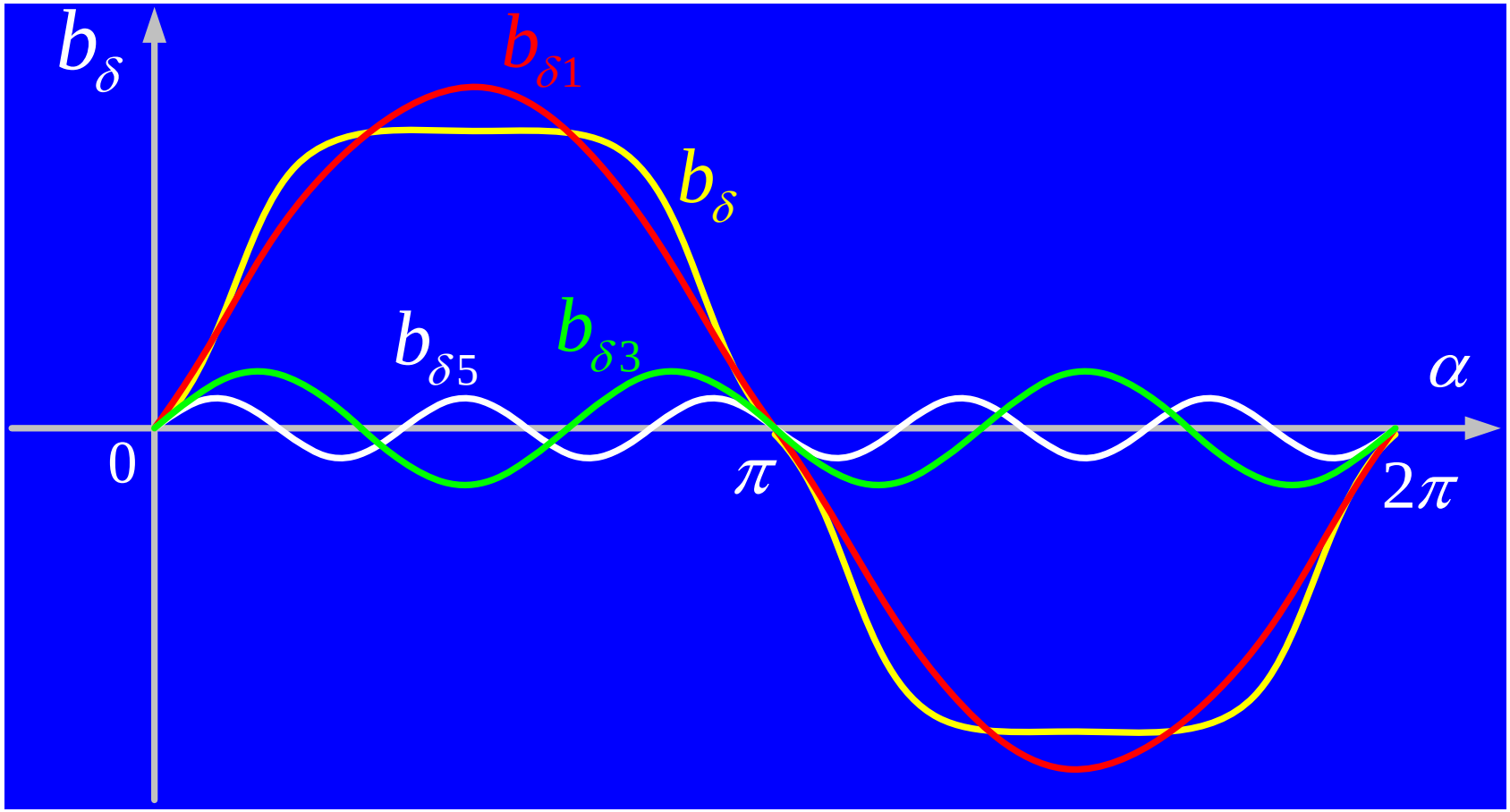






电机气隙磁密波及其分解 ：

$$b_\delta = B_{\delta 1m}\sin a + B_{\delta 3m}\sin 3a + B_{\delta 5m}\sin 5a\cdots$$



作业

- ~~6.1，2，3~~ 能否利用工具得到展开式和频谱
- 6.4，5，6，7，8，11，12，15 非正弦分析
- ~~6.13，14~~ 傅立叶变换与连续频谱
- 6.16，17，18，20 三相非正弦（爱班）
- **6.21，22** 频率特性与滤波器